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Privacy-preserving Sync Architecture for Adaptive Persona RAG System

User device / App

- Local Encrypted Storage
- Raw conversation csv
 - Raw Messages
 - FAISS message index
 - offline Intent model
 - Persona drift json



ON-Device AI Pipeline

1. Ingest message
2. Topic splitter
3. Persona drift detector
4. offline Intent classifier
5. conflict-Aware RAG Resolver



Local Backend + UI

Flask API

Frontend dashboard

APIs:

- /chat
- /Persona-drift
- /Intent
- /resolve-conflict

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↓ Sync only safe summaries
Raw messages stay local

Cloud Sync API

- User authentication
- Device identity
- Version tracking
- Conflict merge service



Cloud metadata store

Stores only

- Topic Summaries
- Checkpoint Summary
- Approved persona facts
- Persona drift labels
- Trigger labels
- Artifact versions

Does not store

- Raw private chats
- Full message embeddings
- Sensitive raw data

Why this design?

- This design keeps private conversations on-device, allows offline AI features, syncs only safe summaries, handles conflicts using versioning & evidence ranking.